

Homework 11 solutions

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7.2

15. We prove this by induction on n . The base case $n = 1$ is trivial as $P(E_1) = P(E_1)$, and for $n = 2$, we have:

$$P(E_1 \cup E_2) = P(E_1) + P((E_1 \cup E_2) \setminus E_1) = P(E_1) + P((E_2 \setminus (E_1 \cap E_2)))$$

which is $\leq P(E_1) + P(E_2)$. Now suppose the statement holds for $n < k$. Let $F = \bigcup_{j < k} E_j$. Then

$$P\left(\bigcup_{j \leq k} E_j\right) = P(F \cup E_k) \leq P(F) + P(E_k) \leq \sum_{j \leq k} P(E_j)$$

which completes the induction.

7.4

4. Let X be a random indicator variable that is 1 if the coin lands heads, and 0 otherwise. Then, $E[X] = 1 \cdot 0.6 = 0.6$. Applying linearity of expectation, the total expected number of heads is $0.6 \cdot 10 = 6$.
8. For each $i \in \{1, 2, 3\}$, let D_i be a random variable for the value of die i . The expected value of a single die roll is $\sum_{i \leq 6} i \cdot \frac{1}{6} = \frac{6(7)}{2} \cdot \frac{1}{6} = \frac{7}{2}$, so for each i , $D_i = \frac{7}{2}$. By linearity of expectation, the expected sum of the dice is the sum of their expected values, so the expected value is $\frac{21}{2}$.
18. First of all, note that for any point $s \in S$, we have $Z(s) \leq X(s) + Y(s)$ since X and Y are always nonnegative. Let $D(s) = (X(s) + Y(s)) - Z(s)$ be a random variable for the difference, so that $Z(s) + D(s) = X(s) + Y(s)$ for all s . Taking the expectation of both sides and applying linearity of expectation, we get $E[Z] + E[D] = E[X] + E[Y]$. Last, note that $E[D]$ is always nonnegative; hence, $E[Z] \leq E[X] + E[Y]$.
24. Recall that the expectation $E[I_A] = \sum_{s \in S} I_A(s) \cdot p(s)$. Since I_A is always 0 or 1, we can split this into $\sum_{s \in A} 1 \cdot p(s) + \sum_{s \notin A} 0 \cdot p(s)$. The right term ends up being 0, so the whole formula is just $\sum_{s \in A} p(s)$, which is just $p(A)$.

26x. We know $E[X] = \sum_{s \in S} X(s)p(s)$. Since X is always a nonnegative integer, we can think of $X(s)p(s)$ as $p(s) + \dots + p(s)$ ($X(s)$ times). So the whole formula becomes

$$\sum_{s \in S} p(s) + \dots + p(s) \text{ (} X(s) \text{ times)}$$

Note that the number of terms inside the sum depends on the value of $X(s)$. Hence, if we distribute our sum above, we get

$$\sum_{s \in S: X(s) \geq 1} p(s) + \sum_{s \in S: X(s) \geq 2} p(s) + \dots = \sum_{s \in A_1} p(s) + \sum_{s \in A_2} p(s) + \dots$$

But also notice that for each k , we have $p(A_k) = \sum_{s \in A_k} p(s)$. Hence, we can rewrite our sum as

$$p(A_1) + p(A_2) + \dots = \sum_{k=1}^{\infty} p(A_k)$$

2016 Practice Midterm question

The total number of unordered pairs of students is $\binom{260}{2}$. For each possible pair of students $\{i, j\}$, let us define an indicator random variable I_{ij} :

$$I_{ij} = \begin{cases} 1 & \text{if student } i \text{ and student } j \text{ have the same birthday} \\ 0 & \text{otherwise} \end{cases}$$

Since the probability that the students will share a birthday is $\frac{1}{365}$, the expected value $E[I_{ij}]$ is just $\frac{1}{365}$.

By linearity of expectation, the expected total is

$$\sum_{i, j \leq 260} E[I_{ij}] = \binom{260}{2} \frac{1}{365} = \frac{260 \cdot 259}{2 \cdot 365} = \frac{130 \cdot 259}{365} \approx 92.25$$

which is quite a bit more than you might have thought before you solved this problem.

2016 Practice Final question

For each bin i , set I_i as a random indicator variable that is 0 if bin i is empty and 1 if bin i is nonempty. Set X as the total number of empty bins, so

$$X = \sum_{i \in \{1, 2, 3, 4\}} I_i$$

Next, we will calculate the probability that bin i is empty. Since the probability of a ball missing bin i is $\frac{3}{4}$, the probability that all 10 balls will miss bin i is $(\frac{3}{4})^{10}$. Thus, the expectation $E[I_i] = (\frac{3}{4})^{10}$ for each bin i .

Applying linearity of expectation gives us that the expected total number of bins is just $4 \cdot (\frac{3}{4})^{10} = \frac{3^{10}}{4^9}$, which is about 0.225.

Nikhil's Problems

1. For each box i , we set an indicator variable Y_i indicating whether box i has exactly one ball. The total value of Y is $\sum_{i \leq n} Y_i$.

To calculate $E[Y_i]$, we just need to calculate the probability that exactly one ball will land in box i . One way to calculate this probability is to consider each specific ball, and find the probability that exactly that ball will land in box i and no others. There are m balls to choose from, and the probability that the ball we choose will land in box i is $\frac{1}{n}$. Then, the probability that all the other balls will miss box i is $(\frac{n-1}{n})^{m-1}$. Putting it all together, we get that the probability that box i will have exactly one ball is

$$m \cdot \frac{1}{n} \left(\frac{n-1}{n} \right)^{m-1} = \frac{m}{n} \left(\frac{n-1}{n} \right)^{m-1}$$

By linearity of expectation, we get

$$E[Y] = n \cdot \frac{m}{n} \left(\frac{n-1}{n} \right)^{m-1} = m \left(\frac{n-1}{n} \right)^{m-1}$$

2. We apply linearity of expectation again.

For each $i \in \{1, \dots, n\}$, set I_i as a random indicator variable indicating whether i is a near-fixed point. To find $E[I_i]$, we just need to calculate the probability that a random permutation will make i a near fixed point.

For $i = 1$, the possible values for $\pi(i)$ to be a near fixed point are 1 and 2. Thus, the probability that 1 is a near fixed point is $\frac{2}{n}$. Similar reasoning applies for $i = n$.

For any other value of i , the possible values for $\pi(i)$ to be a near fixed point are $i-1$, i , and $i+1$. Thus, the probability that $i \in \{2, \dots, n-1\}$ is a near fixed point is $\frac{3}{n}$.

The expected total number of near fixed points is

$$\sum_{i=1}^n E[I_i] = E[I_1] + E[I_n] + \sum_{i=2}^{n-1} E[I_i] = \frac{2}{n} + \frac{2}{n} + (n-2) \left(\frac{3}{n} \right)$$

This simplifies to

$$\frac{4}{n} + \frac{3n-6}{n} = \frac{3n-2}{n}$$

3. Let I_{ij} be a random indicator variable indicating whether $\pi(i) < \pi(j)$. Note that since π is random, it is just as likely that $\pi(i) < \pi(j)$ as $\pi(j) < \pi(i)$. Hence, $E[I_{ij}] = \frac{1}{2}$.

Since there are $\binom{n}{2}$ pairs that are potential inversions, the expected total number of inversions is

$$\binom{n}{2} \cdot \frac{1}{2} = \frac{n(n-1)}{2} \cdot \frac{1}{2} = \frac{n(n-1)}{4}$$

4. For each edge $\{u, v\}$ in the graph, set $I_{\{u, v\}}$ a random indicator variable that indicates whether students u and v were both selected for the committee. Since the students are all selected independently, for any edge $\{u, v\}$ the probability that both u and v were selected is equal to p^2 . Hence, $E[I_{\{u, v\}}] = p^2$.

Applying linearity of expectation, the total expected number of pairs is

$$\sum_{e \in E} E[I_e] = |E| \cdot p^2$$

5. We make a random cut (A, B) by randomly deciding for any vertex v , whether to put v into A or into B .

For each edge e , set I_e a random indicator variable indicating whether e is a crossing edge of our randomly generated cut.

Let $e = \{u, v\}$ be an edge. WLOG assume $u \in A$. The probability that $v \in A$ is $\frac{1}{2}$. Hence, we get $E[I_e] = \frac{1}{2}$. Since there are m total edges, applying linearity of expectation gives us $\frac{m}{2}$ expected total crossing edges.

Since the maximum number of crossing edges cannot be strictly less than the expected number of crossing edges, we have that there is a cut with at least $\frac{m}{2}$ crossing edges.